

ECE 3337: Signals & Systems Analysis

Lecture 14: Laplace Transform

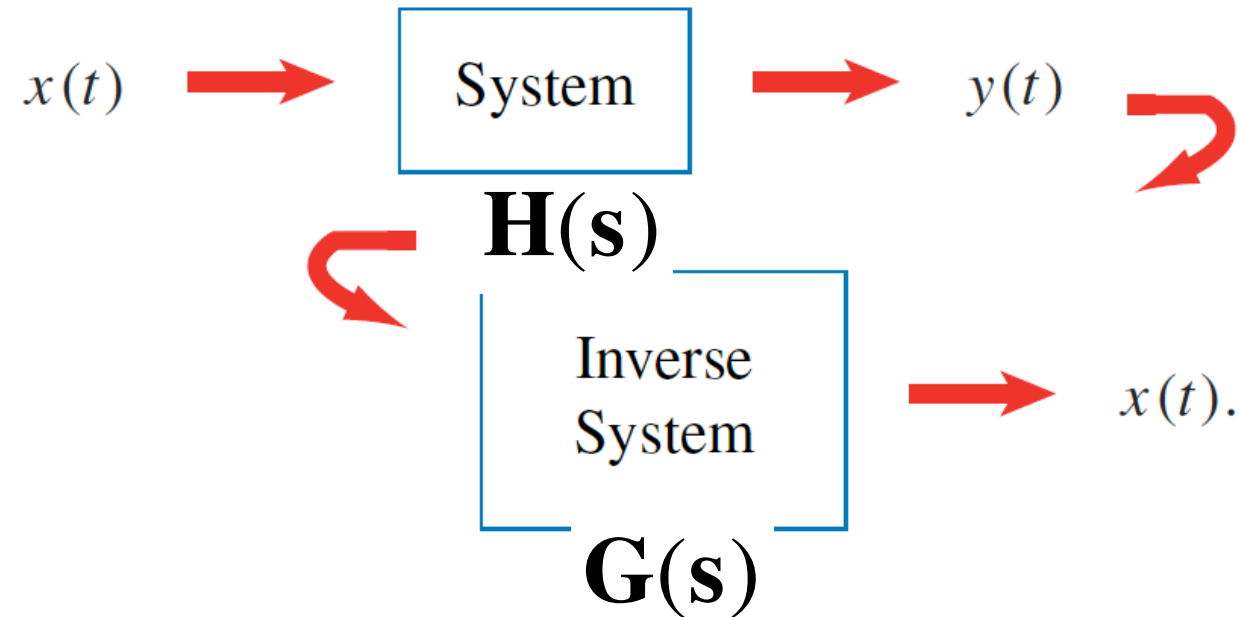
**Reference: Chapter 3
(Sections 3.9 – 3.11)**

Please turn off your gadgets now



Recap: Inversion and Minimum-phase Systems

$$\mathbf{G(s)} = \frac{1}{\mathbf{H(s)}}$$



► A BIBO stable and causal LTI system has a BIBO stable and causal inverse system if and only if all of its poles and zeros are in the open left half-plane, and they are equal in number (its transfer function is proper). Such a system is called a *minimum phase* system. ◀



**DESIRABLE
SYSTEM**



Recap: LCCDE's

Linear constant-coefficient differential equations (LCCDE) occur commonly when describing LTI systems especially, circuits

Example:

$$\frac{d^2 y}{dt^2} + 2 \frac{dy}{dt} + 3y = 4 \frac{dx}{dt} + 5x$$

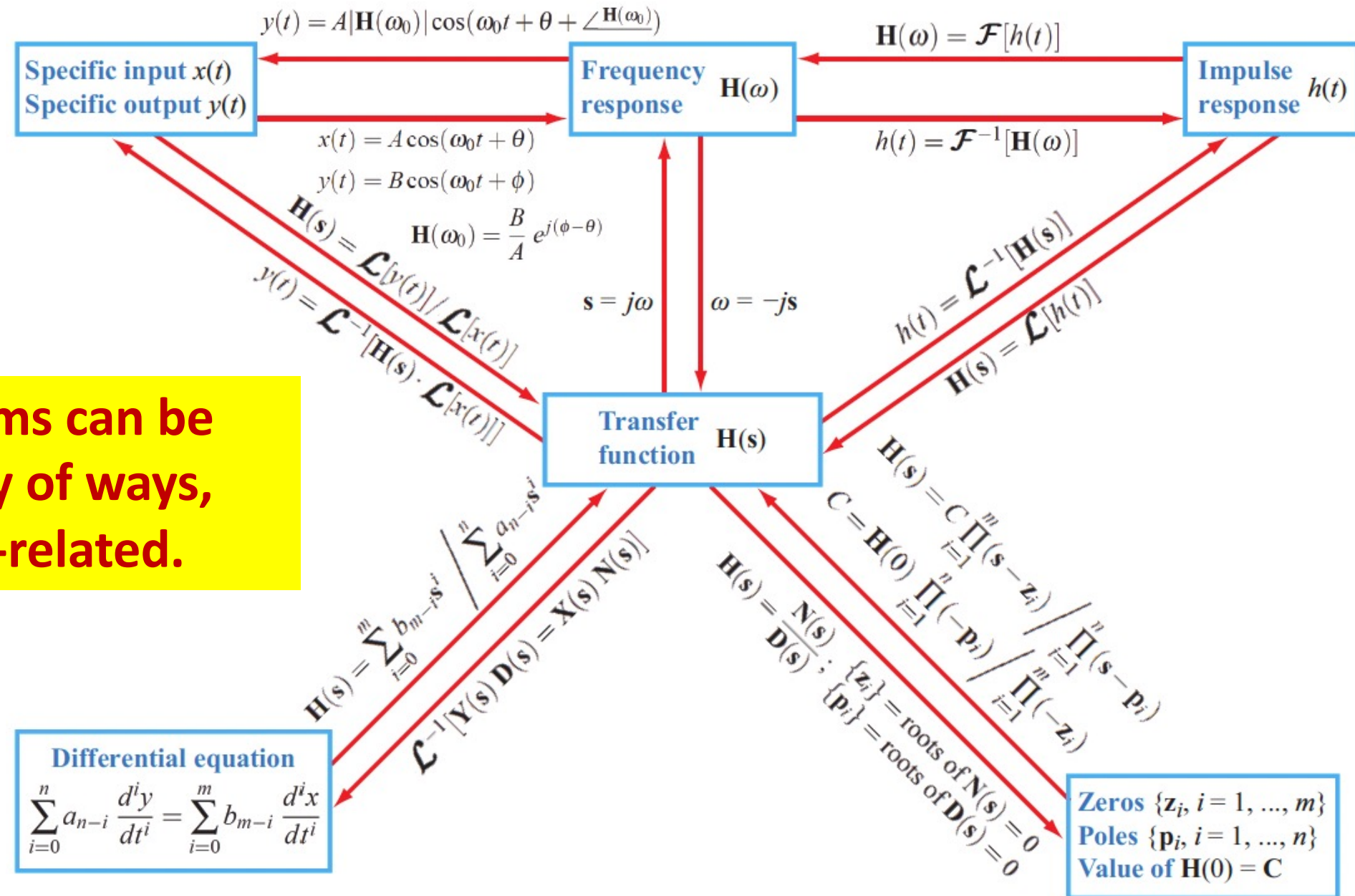
The corresponding **transfer functions** are ratios of polynomials in s

$$\mathbf{H(s)} = \frac{\mathbf{Y(s)}}{\mathbf{X(s)}} = \frac{4s + 5}{s^2 + 2s + 3}$$

A partial fraction expansion of this ratio allows us to find the inverse Laplace transform efficiently



Diagram of LTI Representations



LTI Signals and Systems can be described in a variety of ways, but they are all inter-related.



Today

We will learn to analyze circuits containing resistors, capacitors, and inductors using the Laplace Transform

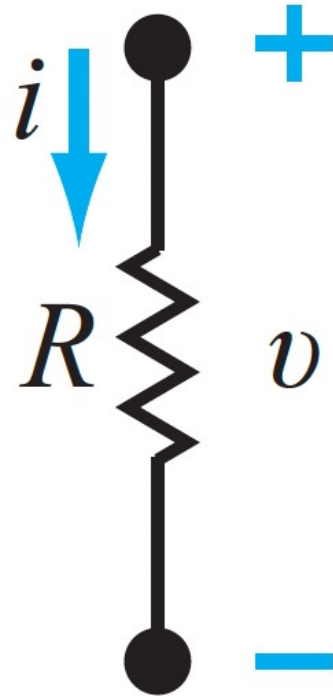
– This is how it is done in real-world practice!

- **We will learn to account for initial conditions in addition to applied inputs**
 - The system's output becomes a mixture of responses
 - We need to learn how to make sense of it
- **We will also learn to parse the system's output “mixture” into its constituent parts**

• Zero-input Response	and	Zero-state Response
• Natural Response	and	Forced Response
• Transient Response	and	Steady-state Response

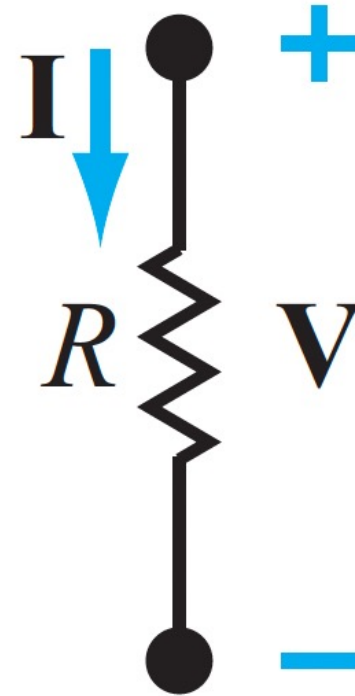


Resistors



Time Domain

$$v(t) = R i(t)$$



$$V(s) = R I(s)$$

Laplace Domain

Resistors cannot store any energy, so no initial conditions are involved



Capacitors

VOLTAGE DIFFERENTIATOR

$$i_C = C \frac{dv_C}{dt}$$



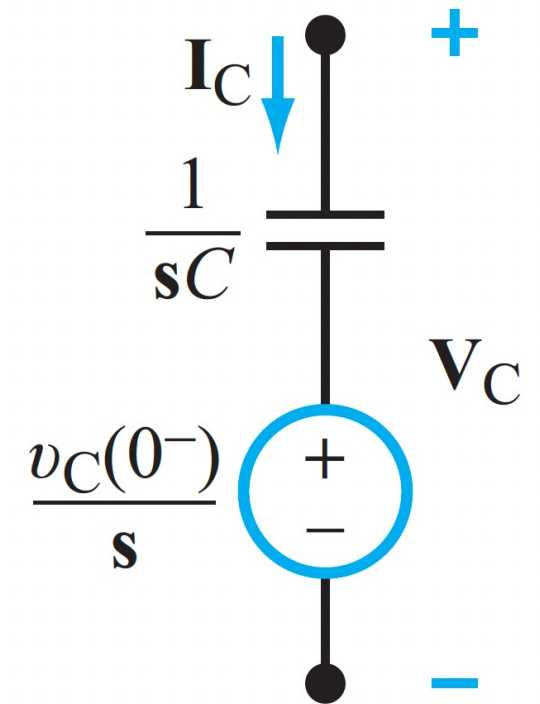
$$\mathbf{I}_C(s) = C[s\mathbf{V}_C(s) - v_C(0^-)]$$

$$v_C = \frac{1}{C} \int_{0^-}^t i_C dt + v_C(0^-)$$



$$\mathbf{V}_C(s) = \frac{1}{C} \left[\frac{\mathbf{I}_C(s)}{s} \right] + \frac{v_C(0^-)}{s}$$

CURRENT INTEGRATOR



Capacitors can have an initial voltage $v_C(0^-)$.
The initial voltage behaves like a step function in series



Inductors

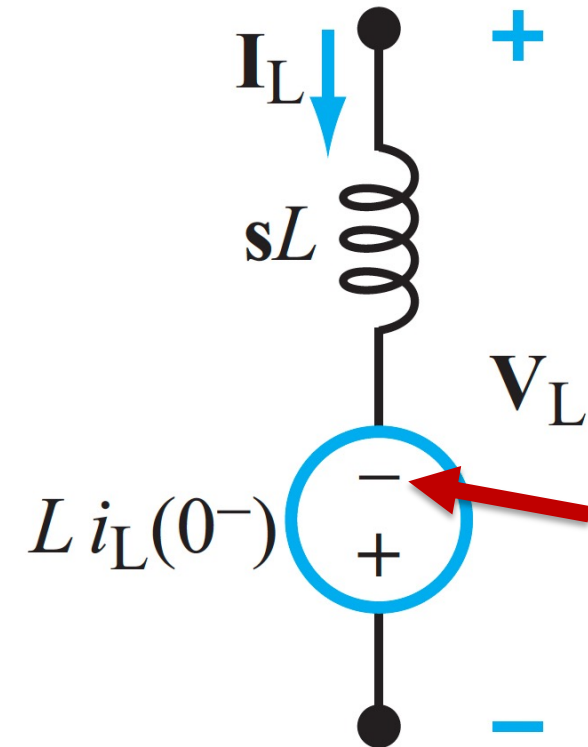
$$v_L = L \frac{di_L}{dt} \quad \text{CURRENT DIFFERENTIATOR}$$

$$i_L = \frac{1}{L} \int_{0^-}^t v_L dt + i_L(0^-)$$

VOLTAGE INTEGRATOR

$$\mathbf{I}_L(s) = \frac{1}{sL} \mathbf{V}_L(s) + \frac{i_L(0^-)}{s}$$

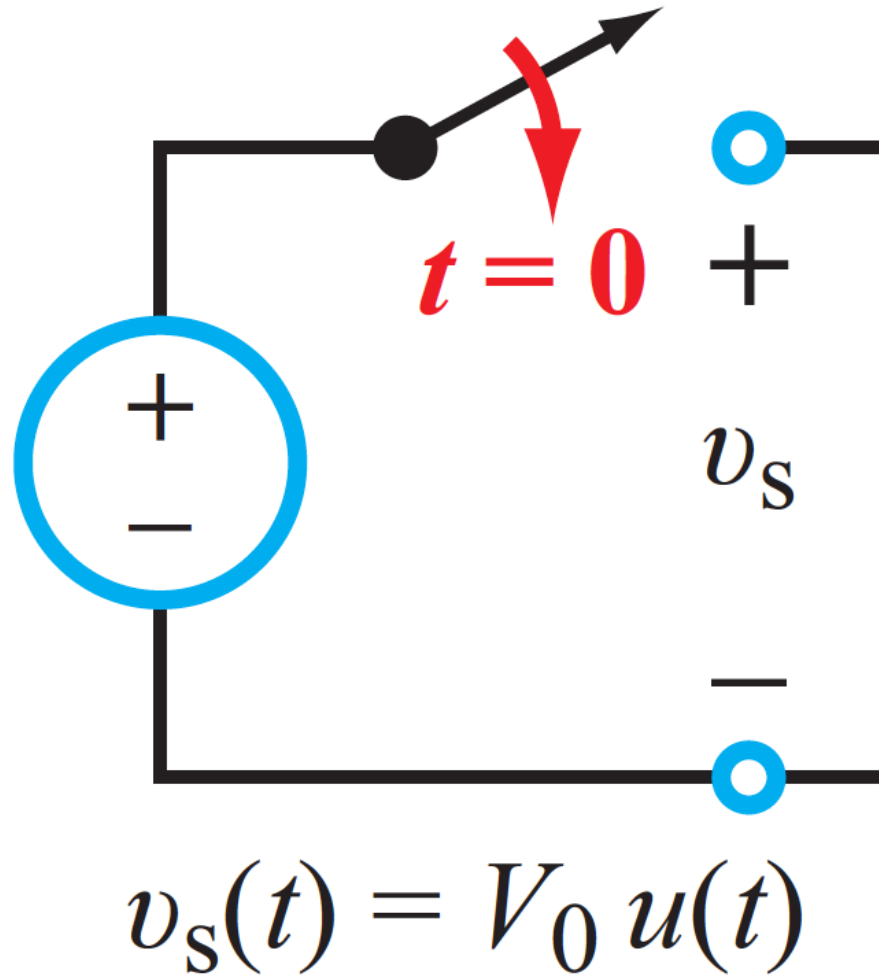
$$\mathbf{V}_L(s) = L[s\mathbf{I}_L(s) - i_L(0^-)]$$



Inductors can have an initial current
Manifests as a negative voltage in series



Voltage Source

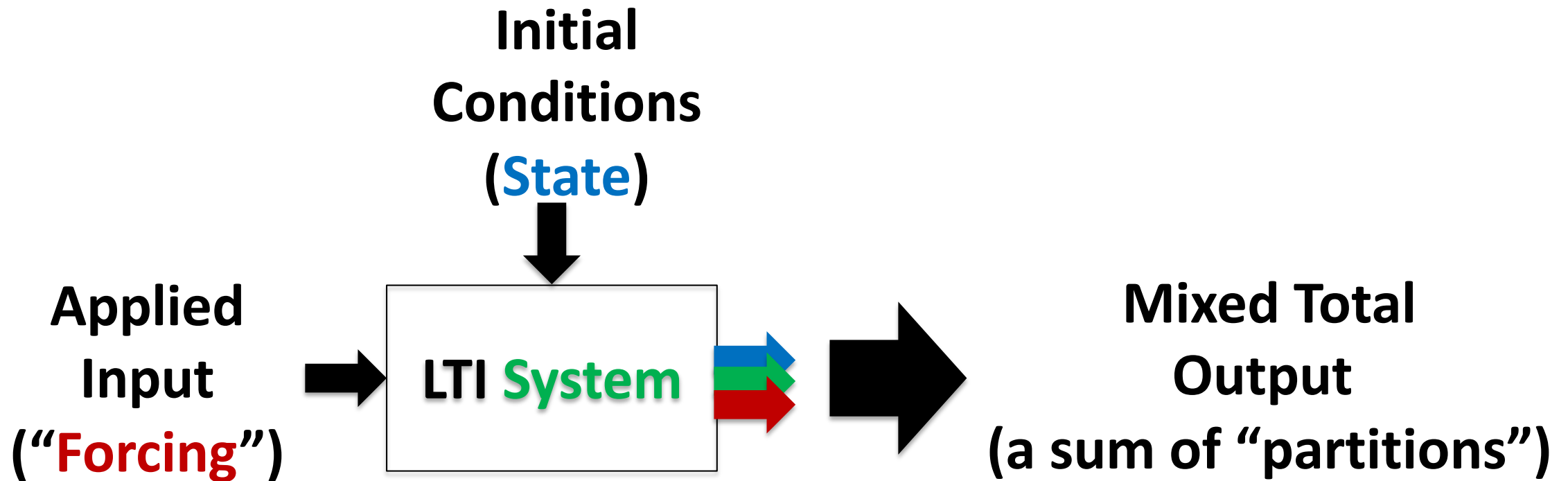


$$\mathcal{L}[V_0 u(t)] = V_0 \frac{1}{s}$$

STEP FUNCTION APPLIED AT $t = 0$



Today's Big Question

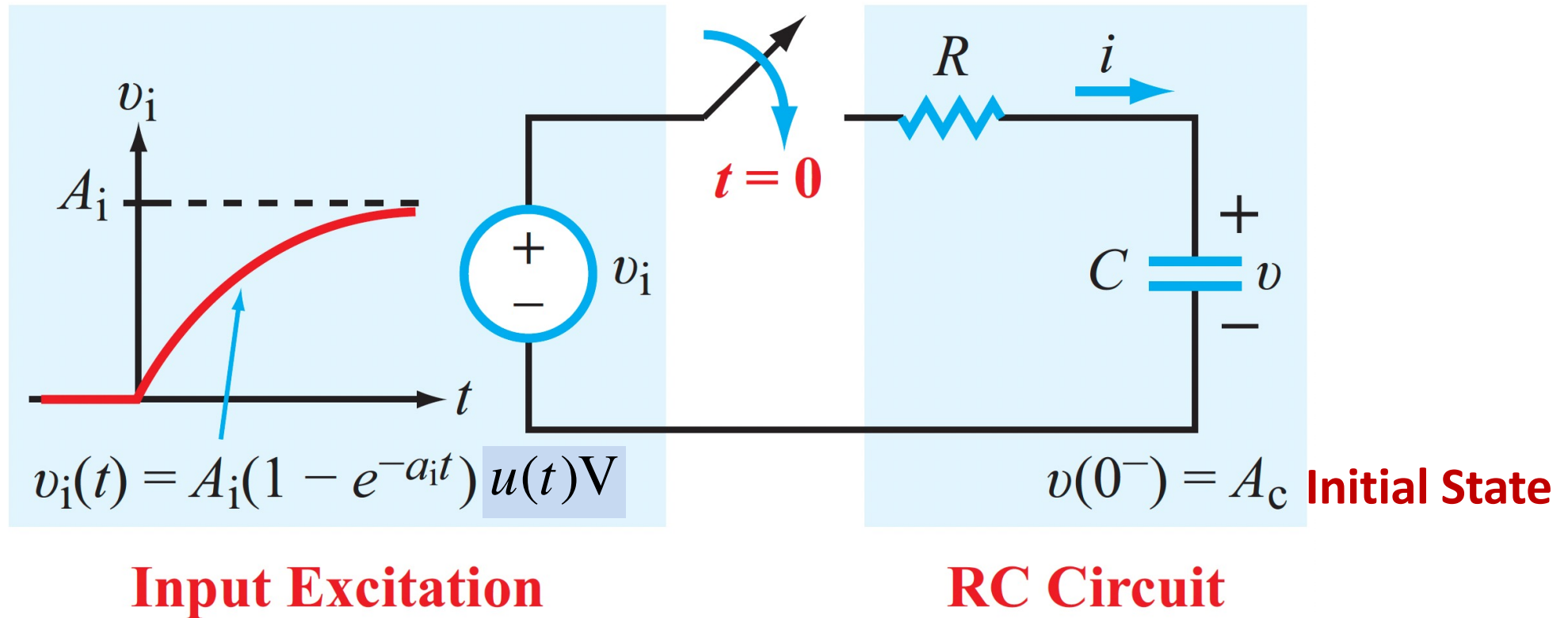


The system's output contains contributions from the input, the system's transfer function, and initial conditions

Can we “untangle” the output mixture into its partitions?



RC Circuit Example

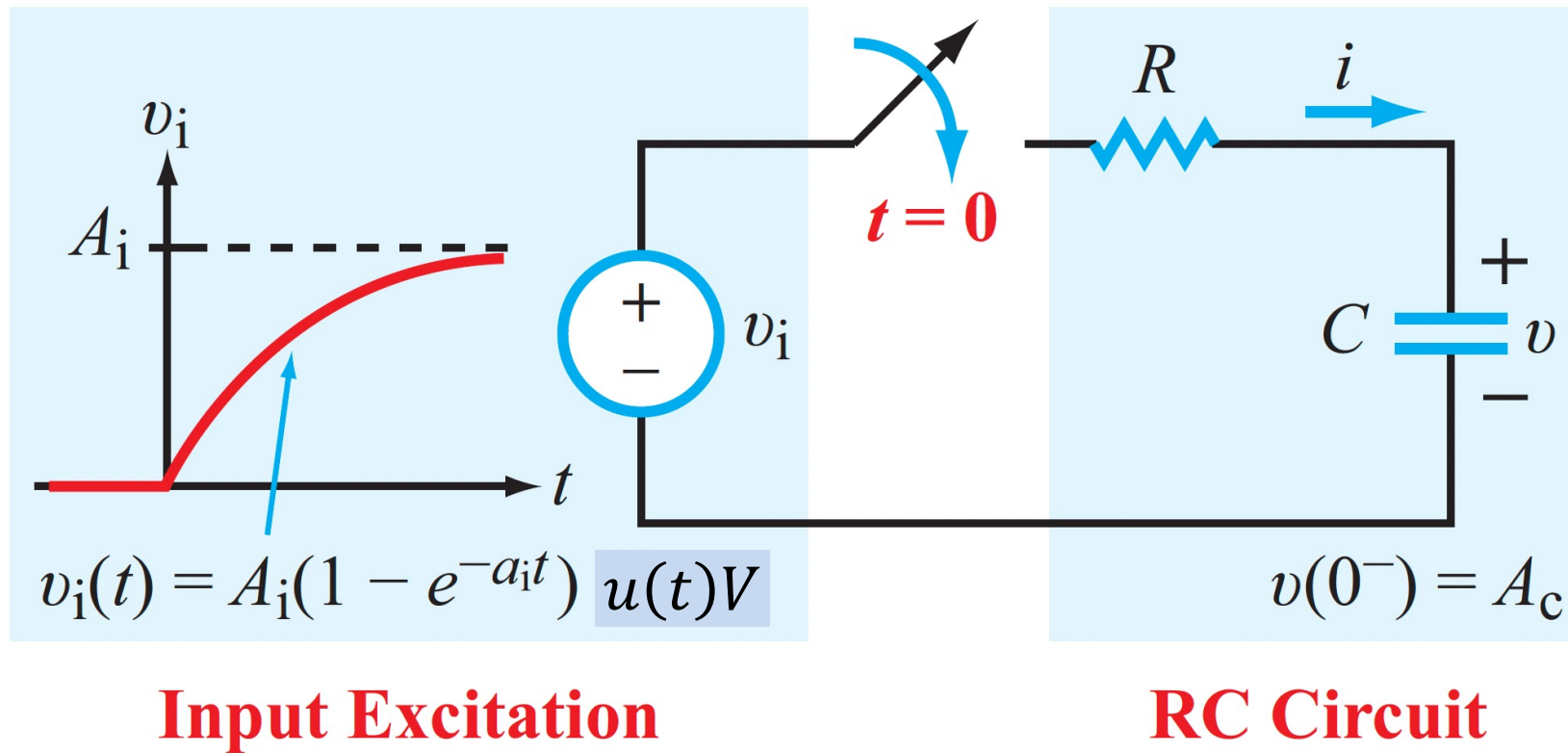


Notation for Today:

- We use the subscript c to label anything that is only related to the circuit (a_c, A_c)
- We use the subscript i to label anything that pertains to the input (v_i, a_i, A_i)



Applying KVL



Applying KVL at

$$t \geq 0$$

$$v_i(t) = i(t)R + v$$

$$= RC \frac{dv(t)}{dt} + v(t)$$



Taking the Laplace Transform

$$v_i(t) = RC \frac{dv(t)}{dt} + v(t)$$

$$v(t) \rightarrow \mathbf{V}(s)$$

$$\frac{dv(t)}{dt} \rightarrow s\mathbf{V}(s) - v(0^-) \quad \text{Property 6}$$

$$v_i(t) \rightarrow \mathbf{V}_i(s)$$

$$\Rightarrow \mathbf{V}_i(s) = RC \left[s\mathbf{V}(s) - v(0^-) \right] + \mathbf{V}(s)$$



Solving for the Output

$$\mathbf{V}_i(\mathbf{s}) = RC \left[\mathbf{sV}(\mathbf{s}) - v(0^-) \right] + \mathbf{V}(\mathbf{s})$$

$$\mathbf{V}_i(\mathbf{s}) + RCv(0^-) = \mathbf{V}(\mathbf{s})[sRC + 1]$$

$$\Rightarrow \mathbf{V}(\mathbf{s}) = \frac{RC}{sRC + 1} v(0^-) + \frac{1}{sRC + 1} \mathbf{V}_i(\mathbf{s})$$



Transfer Function

The transfer function of the circuit $H(s)$ is the ratio of the Laplace Transform of the output $V(s)$ to the Laplace transform of the input $V_i(s)$

- Important requirement for $H(s)$: All initial conditions should be 0 at $t = 0$ –

$$H(s) = \frac{V(s)}{V_i(s)} = \frac{1}{sRC + 1}$$



Interpreting the Output

The system's output can be parsed and interpreted in a variety of useful ways, for example:

$$\underbrace{V(s)}_{\text{TOTAL RESPONSE}} = \underbrace{\frac{RC}{sRC + 1} \overset{\text{STATE}}{v(0^-)}}_{\text{ZERO-INPUT RESPONSE}} + \underbrace{\frac{1}{sRC + 1} \overset{\text{INPUT}}{V_i(s)}}_{\text{ZERO-STATE RESPONSE}}$$



Definitions

Zero-input Response (ZIR): The response of the system in the absence of an input excitation, i.e.,

$$\mathbf{V}_i(\mathbf{s}) = 0$$

Zero-state Response (ZSR): The response of a system when it is in zero state

- zero initial conditions
- zero capacitor voltages and inductor currents



Simplify the Notation

$$\mathbf{V}(s) = \frac{RC}{sRC + 1} v(0^-) + \frac{1}{sRC + 1} \mathbf{V}_i(s)$$

$$\text{Let } RC = \frac{1}{a_c}, \text{ and } v(0^-) = A_c$$

$$\Rightarrow \mathbf{V}(s) = \frac{A_c}{s + a_c} + \frac{a_c}{s + a_c} \mathbf{V}_i(s)$$

**TOTAL
RESPONSE**

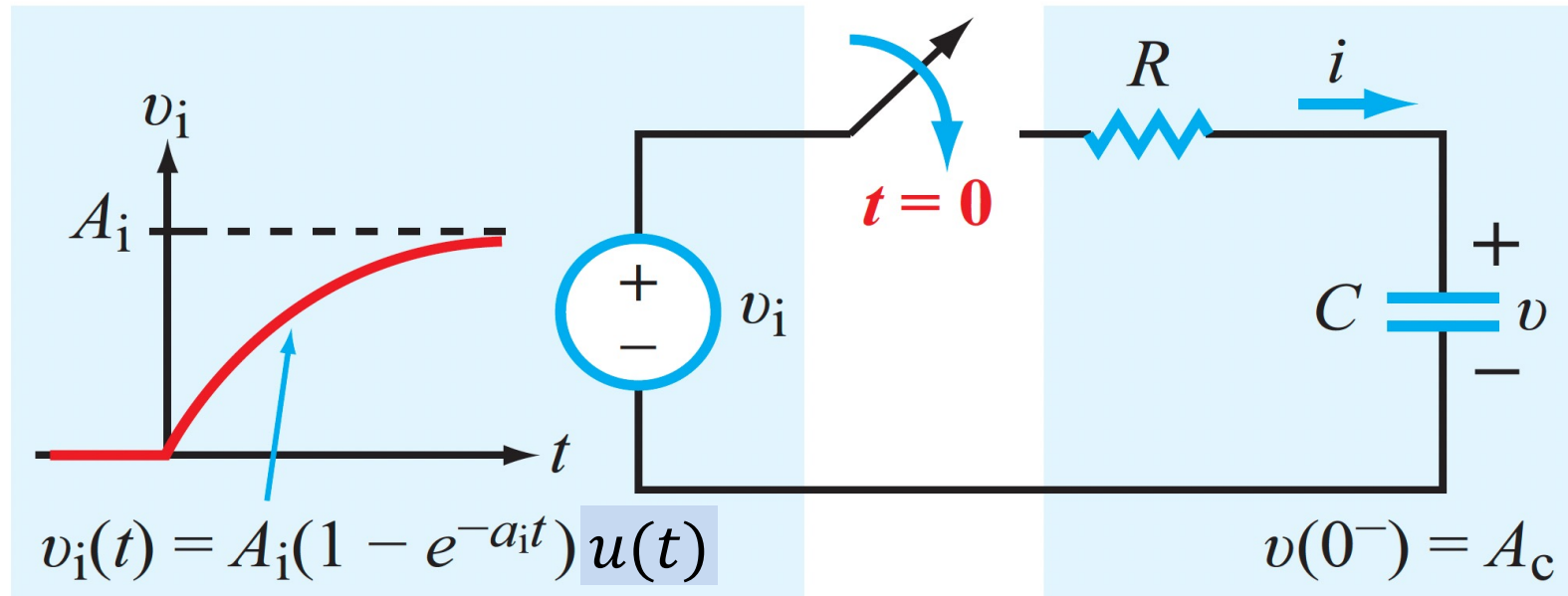
**ZERO-INPUT
RESPONSE**

**ZERO-STATE
RESPONSE**



Transforming the Input

Now, let's Laplace transform the input



Input Excitation

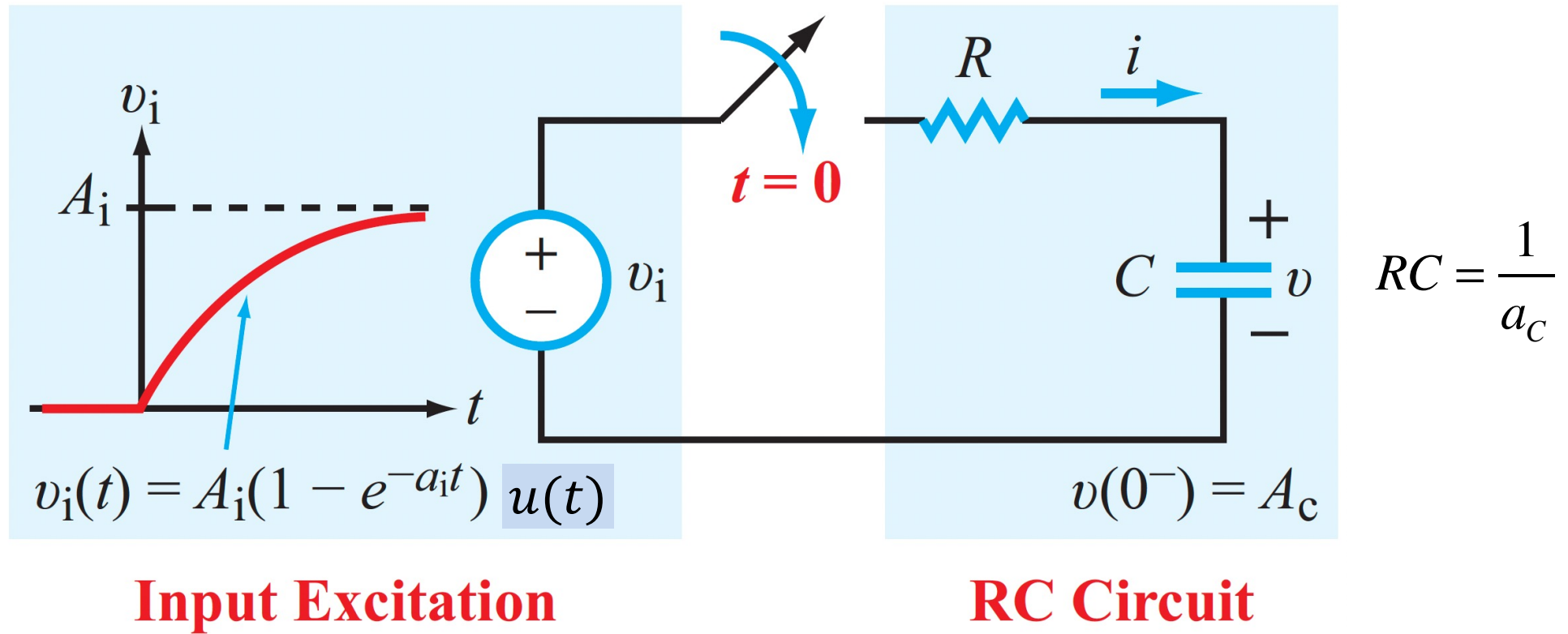
RC Circuit

$$v_i(t) = A_i(1 - e^{-a_i t})u(t)$$

$$\mathbf{V}_i(s) = A_i \left[\frac{1}{s} - \frac{1}{(s + a_i)} \right] = \frac{A_i a_i}{s(s + a_i)}$$



Transforming the Output



$$\Rightarrow \mathbf{V(s)} = \frac{A_c}{s + a_c} + \frac{a_c}{(s + a_c)} \left[\frac{A_i a_i}{s(s + a_i)} \right]$$

PARTIAL FRACTION EXPANSION NEEDED



Recap: Partial Fractions

By now, you may have noticed that there is a simple pattern to partial fraction expansions with first-order terms.

Example:

$$\frac{as + b}{(s + c)(s + d)} = \frac{a(-c) + b}{(s + c)(-c + d)} + \frac{a(-d) + b}{(-d + c)(s + d)}$$

This is the residue method for partial fractions in action.



Partial Fraction Expansion

$$V(s) = \frac{A_c}{s + a_c} + \frac{a_c}{(s + a_c)} \left[\frac{A_i a_i}{s(s + a_i)} \right]$$

PARTIAL FRACTION EXPANSION

$$= \underbrace{\frac{A_c}{s + a_c}}_{\text{ZERO-INPUT RESPONSE}} + \underbrace{\left[\frac{A_i}{s} + \frac{A_i a_i}{(a_c - a_i)} \left(\frac{1}{\underline{s + a_c}} \right) + \frac{A_i a_c}{(a_i - a_c)} \left(\frac{1}{\underline{s + a_i}} \right) \right]}_{\text{ZERO-STATE RESPONSE}}$$

**ZERO-INPUT
RESPONSE**

**ZERO-STATE
RESPONSE**

DEPENDS ON CIRCUIT COMPONENTS ALONE

DEPENDS ON THE INPUT, AND CIRCUIT COMPONENTS



Convert to Time Domain

$$\mathbf{V}(s) = \frac{A_c}{s + a_c} + \left[\frac{A_i}{s} + \frac{A_i a_i}{(a_c - a_i)} \left(\frac{1}{s + a_c} \right) + \frac{A_i a_c}{(a_i - a_c)} \left(\frac{1}{s + a_i} \right) \right]$$



$$v(t) = A_c e^{-a_c t} u(t) + \left[A_i + \frac{A_i a_i}{(a_c - a_i)} e^{-a_c t} + \frac{A_i a_c}{(a_i - a_c)} e^{-a_i t} \right] u(t)$$

The term $e^{-a_c t}$ is called a "mode" of the system

$$a_c = \frac{1}{RC}$$

**MODES DEPEND ON CIRCUIT COMPONENTS ALONE
THEY DEFINE THE NATURAL CHARACTER OF THE SYSTEM**



Modes of a System

Definition: The modes of an LTI system are the roots of the characteristic polynomial

- They come from the coefficients of the left-hand side of the LCCDE
- The modes define the system's natural behavior

Example:

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = x(t) + 2 \frac{dx(t)}{dt}$$

The characteristic polynomial is $s^2 + 3s + 2 = (s + 1)(s + 2)$

The modes are $\{-1, -2\}$. They manifest as poles at $\{-1, -2\}$ in the s-domain, and as exponential terms e^{-t} and e^{-2t} in the time-domain



Another way to Partition

We can also partition the same output based on the concept of natural/system “modes”

$$v(t) = A_C e^{-a_C t} u(t) + \left[A_i + \frac{A_i a_i}{(a_C - a_i)} e^{-a_C t} + \frac{A_i a_C}{(a_i - a_C)} e^{-a_i t} \right] u(t)$$
$$= \underbrace{\left[A_C + \frac{A_i a_i}{(a_C - a_i)} \right] e^{-a_C t} u(t)}_{\text{NATURAL RESPONSE}} + \underbrace{\left[A_i + \frac{A_i a_C}{(a_i - a_C)} e^{-a_i t} \right] u(t)}_{\text{FORCED RESPONSE}}$$

**NATURAL
RESPONSE**

**FORCED
RESPONSE**

$$RC = \frac{1}{a_C}$$

**ALL THE TERMS THAT
DEPEND ONLY ON THE
SYSTEM'S MODES**

ALL REMAINING TERMS



Yet Another way to Partition

We can also partition the output based on how the components of the output behave over time

$$v(t) = A_C e^{-a_C t} u(t) + \left[A_i + \frac{A_i a_i}{(a_C - a_i)} e^{-a_C t} + \frac{A_i a_C}{(a_i - a_C)} e^{-a_i t} \right] u(t)$$
$$= \underbrace{\left[A_C e^{-a_C t} + \frac{A_i a_i}{(a_C - a_i)} e^{-a_C t} + \frac{A_i a_C}{(a_i - a_C)} e^{-a_i t} \right] u(t)}_{\text{TRANSIENT RESPONSE}} + \underbrace{A_i u(t)}_{\text{STEADY-STATE RESPONSE}}$$

**TRANSIENT
RESPONSE**

**ALL TERMS THAT DECAY TO
ZERO OVER TIME**

**STEADY-STATE
RESPONSE**

**TERMS THAT REMAIN
AFTER THE TRANSIENTS
GO TO ZERO**



In the Real World

- **The transient response is important in engineering design**
 - We need to ensure that our system has an acceptable transient response:
 - Under-damped / Over-damped / Critically Damped
 - Acceptable rise time, overshoot, settling time
 - For example, transients should not damage components and circuits!
- **With the steady-state response, we need to ensure that it is not too far from the expected settling point**
 - Minimum steady-state error
 - We'll discuss this further in the next class



Step-by-step Process with LCCDE

Given a system and an input, we go about the analysis as follows:

- The system can be specified in a variety of ways (e.g., circuit, LCCDE, etc.)
- The input can be specified in the time domain or s domain

Step 1: Convert everything to the s domain. If given a circuit, you must solve the circuit using KVL/KCL. This will result in an equation containing all initial conditions, for example:

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} + 2x(t)$$

$$[s^2 Y(s) - sy(0-) - \frac{dy(0-)}{dt}] + 3[sY(s) - y(0-)] + 2Y(s)$$

$$= [sX(s) - x(0-)] + 2X(s)$$



Step-by-step Process

Step 2: To calculate the transfer function, set all initial conditions to zero and solve for $Y(s)/X(s)$

$$[s^2 Y(s)] + 3[sY(s)] + 2Y(s) = [sX(s)] + 2X(s)$$

$$\Rightarrow H(s) = \frac{Y(s)}{X(s)} = \frac{s + 2}{s^2 + 3s + 2}$$

Step 3: The zero-state response is calculated by setting all initial conditions to 0 and substituting the Laplace transform for the given input $X(s)$

$$Y_{ZSR}(s) = H(s)X(s) = \frac{s + 2}{s^2 + 3s + 2} X(s)$$



Step-by-step Process

Step 4: To calculate the zero-input response, set the input to zero in the original equation and solve for the output

$$[s^2 Y(s) - sy(0-) - \frac{dy(0-)}{dt}] + 3[sY(s) - y(0-)] + 2Y(s)$$

$$= [sX(s) - x(0-)] + 2X(s) = 0$$

$$\Rightarrow Y_{ZIR}(s) = \frac{sy(0-) + \frac{dy(0-)}{dt} + 3y(0-)}{s^2 + 3s + 2}$$



Step-by-step Process

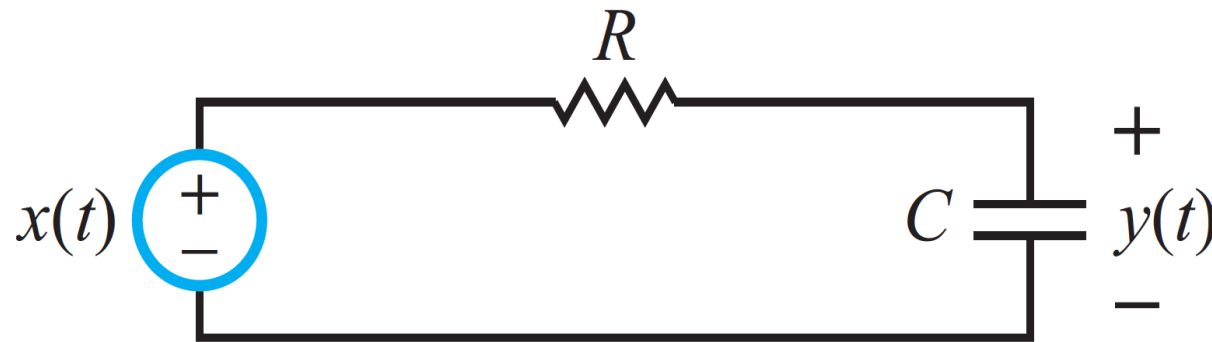
Step 5: To calculate the rest of the response partitions, plug in $\mathbf{X(s)}$ and solve for the full output. This is simply given by adding the two responses we just calculated:

$$\mathbf{Y(s)} = \mathbf{Y_{ZSR}(s)} + \mathbf{Y_{ZIR}(s)}$$

- Do the partial fraction expansion of this full response
- Collect all terms whose poles correspond to the poles of $H(s)$. This sum is the **natural response** of the system. All other terms are the **forced response**.
- Collect all terms with decaying exponentials. This sum is the **transient response** of the system. All other terms are the **steady-state response**.



Activity: Please Attempt This Problem



$$x(t) = 25 \cos(3t)u(t)$$

$$R = 250k\Omega$$

$$C = 1\mu F$$

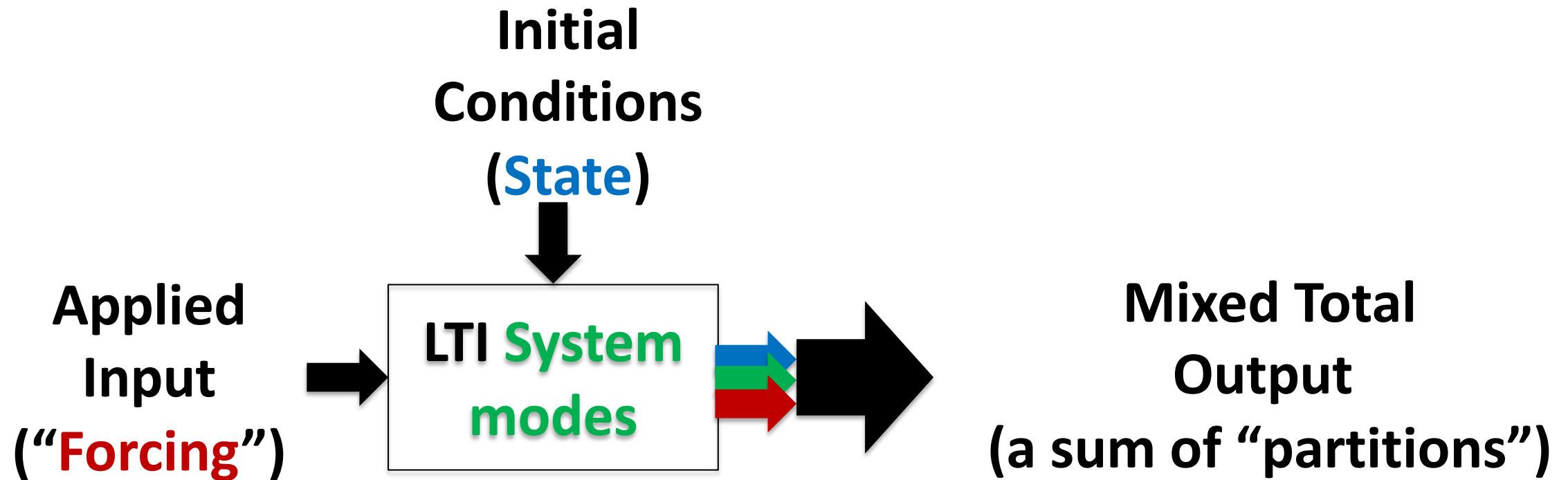
$$v_C(0-) = 2V$$

CALCULATE THE FOLLOWING: Zero-input response.
Zero-state response.
Transient response.
Steady-state response.
Natural response.
Forced response.

**DETAILED
SOLUTION
POSTED ON
CANVAS SITE**



Summary



We learned to parse a system's output into its partitions:

Zero-input Response

and

Zero-state Response

Natural Response

and

Forced Response

Transient Response

and

Steady-state Response



For the Next Class

Work through the textbook examples in Section 3-11

Study Sections 4.1 – 4.2

How to Study:

- 1. Read the assigned sections to understand the basic concepts, and aim to memorize the concepts**
- 2. Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
- 3. Write down your questions and bring them to class**